DEEDY: A Thai-Centric Multi-Agent Framework for Social-Media Marketing Campaign Analysis

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***Abstract*—Social-media marketing campaigns generate large volumes of fast-moving, context-dependent consumer reactions that are difficult to interpret with sentiment classification alone. This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), a Thai-centric application and experimental testbed for analyzing campaign response with large-language-model (LLM) agents. For each campaign post, an Apify-based pipeline requests up to 1,000 publicly accessible comments from Facebook, Instagram, or TikTok, preserves post and collection provenance, and produces synchronized raw and Thai-aware analysis views. Campaign facts and approved context form a reality seed, whereas observed reactions reserved for evaluation remain hidden from the agents. DEEDY adapts MiroFish, an open-source multi-agent prediction engine that converts seed documents into a GraphRAG-grounded world, generates actors, runs OASIS social interactions, and supports report generation and post-run inspection. The resulting application lets marketers examine how message framing, audience composition, and platform mechanisms may shape simulated reaction, diffusion, and narrative formation. We propose leakage-resistant evaluation against held-out Thai comments using sentiment and stance distribution distance, narrative coverage, response diversity, propagation error, repeated runs, and native-Thai human assessment. DEEDY is positioned as an exploratory campaign-planning and social-listening instrument, not as a deterministic forecast of consumer behavior or a replacement for field research.**

***Index Terms*—LLM agents, social-media marketing, campaign analysis, social simulation, Thai NLP, GraphRAG, MiroFish, Apify**

I. Introduction

Social-media campaigns develop through interaction: consumers respond to a post, influence one another, and amplify or contest emerging narratives. Engagement counts and independent sentiment labels summarize reaction but do not represent how exposure, replies, and recommendation produce collective outcomes. Agent-based modeling (ABM) instead models local decisions and their emergent effects [1].

LLM agents add profile-, memory-, and context-conditioned language and actions to ABM. Prior systems demonstrate believable agents, populated social prototypes, and large-scale networked interaction [2], [3], [4], [5], [6]. This creates an opportunity to rehearse campaign framing and platform conditions as hypotheses for subsequent testing.

Thai campaign analysis nevertheless requires language- and culture-aware modeling of informal spelling, code-mixing, particles, emoji, and indirect or sarcastic responses [7], [8]. Collection is also platform-dependent, so every result must retain its post, query limit, timestamp, and visibility constraints. Finally, plausible generated comments do not establish behavioral validity; authentic reactions must remain held out and stochastic simulations must be repeated [9], [10].

This paper presents **DEEDY**, a Thai-centric application for social-media marketing campaign analysis. It uses Apify to request up to 1,000 public comments per post from Facebook, Instagram, or TikTok, adapts the MiroFish multi-agent engine [11], and evaluates simulated campaign response against unseen observations. Analysts can compare message framing, audience composition, networks, or feed policies, but the results are treated as exploratory evidence rather than deterministic market forecasts.

Our contributions are threefold:

- An Apify pipeline for collecting up to 1,000 comments per campaign post with normalized fields, provenance, and Thai-aware data views.
- A campaign-focused adaptation of MiroFish/OASIS with Thai actors, configurable interventions, and auditable simulation outputs.
- A leakage-resistant evaluation against held-out Thai reactions using repeated runs, ablations, native-Thai review, and uncertainty reporting.

II. Related Work

A. Generative Agents and Social Simulation

Traditional ABMs encode actor states and behavioral rules explicitly. LLM agents complement this approach by generating context-dependent language and actions, but introduce sensitivity to prompts, decoding, memory, and the shared base model. Generative Agents use observation, reflection, and planning to produce coherent-seeming routines [2]. S^3 connects LLM perception and action to emotion, attitude, and a social network [4]. OASIS adds dynamic user and post state, follow/comment/repost actions, and configurable interest- and popularity-based recommendation at large agent scales [5]. GenSim emphasizes reusable scenario functions and execution robustness [6]. DEEDY uses these systems as infrastructure, but focuses on Thai campaign construction and empirical validation rather than maximum population size.

B. Grounding and Operational Validity

MiroFish organizes document-grounded simulation into five stages: GraphRAG construction from uploaded reality seeds;

entity, persona, and environment setup; OASIS-based social simulation with temporal memory updates; report generation; and post-run interaction with agents or the report agent [11]. This workflow accelerates scenario construction and is therefore the application foundation adapted by DEEDY. However, a MiroFish report is evidence about the simulated world, not evidence that the world is behaviorally valid. DEEDY consequently compares its traces with unseen observations.

Algorithmic fidelity compares LLM-generated and human subgroup distributions [12]. Conditioned Comment Prediction instead compares a generated response with the authentic response a user left to a stimulus, providing a closer operational test for social-media simulation [10]. Validation of generative ABMs further benefits from staged replication before novel counterfactual tests [13]. These ideas motivate DEEDY's campaign-level holdout: material required to describe the campaign is available to agents, whereas the consumer reactions used for scoring are not.

C. Thai Online Language and Campaign Response

Thai NLP remains comparatively resource-constrained, especially for informal and downstream semantic tasks [7]. PyThaiNLP provides tokenization and other open Thai processing components [14]; WangchanBERTa showed that Thai-specific preprocessing and pretraining over news and social data matter [15]; and Typhoon 2 extends Thai-optimized models to instruction-following generation [16]. ThaiCLI separately evaluates cultural and linguistic knowledge, highlighting that general benchmark strength does not guarantee Thai cultural competence [8].

Online affect is also contextual. Sarcasm can invert literal polarity, and conversation history and user style improve its detection [17]. This is important in Thai discussions, where particles, quotation, exaggerated praise, wordplay, and shared cultural references may signal stance indirectly. Accordingly, DEEDY keeps thread context and raw text for evaluation and does not treat sentiment, stance, and sarcasm as interchangeable labels.

III. DEEDY Framework

A. System Architecture Overview

Figure 1 organizes DEEDY into three layers. The *data layer* selects campaign posts, collects comments through Apify, cleans, partitions, and annotates Thai campaign material. The *core layer* supplies the scenario summary to MiroFish, which constructs a GraphRAG-backed world, creates heterogeneous actors, executes social interactions, and generates a simulation report. The *validation layer* compares outputs with social posts and comments never shown to the agents. Every derived artifact records a campaign identifier, platform, post URL, provenance pointer, processing version, and experimental split so that a score can be traced back to its evidence.

B. Data Preparation Pipeline

1) Campaign Definition and Post Selection: The input manifest `data/topic_ref.json` defines one marketing campaign per record through a campaign topic and a list of reference URLs. URLs may refer to campaign posts, creator or influencer posts, news coverage, or public community posts on Facebook, Instagram, and TikTok. Including more than the brand's official account reduces the risk of measuring only an owned-audience response. For an experimental release, the manifest is extended with a stable campaign identifier, observation window, and source stratum. Each URL is manually checked for campaign relevance and resolved to a canonical post before collection. The sampling frame and inclusion rules are frozen before labels or simulation outputs are inspected.

Campaign facts used to construct the simulated scenario are stored separately from consumer reactions. The factual packet may include the campaign brief, creative copy, product information, publication time, and dated public reporting. It does not include the target comments later used for evaluation. This distinction supports both retrospective campaign analysis and a stricter prospective design in which only information available at campaign launch is provided to the agents.

2) Apify Comment Collection: DEEDY uses Apify Actors as the collection interface rather than a manually maintained browser crawler. An Actor is a server-side program that accepts structured input, executes as a recorded run, and writes results to an Apify dataset accessible through the API or client libraries [18]. The implemented `apify_crawler.py` detects the platform from each URL and invokes `apify/facebook-comments-scraper`, `apify/instagram-comment-scraper`, or `clockworks/tiktok-comments-scraper`. A platform adapter maps each Actor's output into one common comment schema.

The collection unit is one social-media post. Under the revised protocol, DEEDY launches an independent Actor query for every canonical post URL with a requested limit of **1,000 comments per post**. The limit is reset for each URL; it is not a shared 1,000-comment budget for the campaign or topic. If a campaign contains m posts, the intended maximum is therefore

$$N_{\max} = 1{,}000m. \tag{1}$$

The returned count may be lower when a post has fewer comments or when the platform or Actor cannot expose every comment. Replies are configured explicitly for each platform and are counted as separate records when returned. Accordingly, "1,000" is a reproducible query cap, not a guarantee of 1,000 observations and not a claim of complete platform coverage.

3) Normalization and Provenance: Each returned item is validated and normalized to the fields `topic`, `platform`, `post_url`, `comment_id`, anonymized `author`, `text`, `likes_count`, `parent_comment_id`, and `timestamp`. The parent identifier preserves reply structure when the Actor exposes it. Missing values remain explicit rather than being inferred. Records without comment text are excluded from content analysis but retained in run-level quality-control counts when they represent Actor errors.

For reproducibility, a provenance manifest records the campaign and canonical post identifiers, Actor identifier and

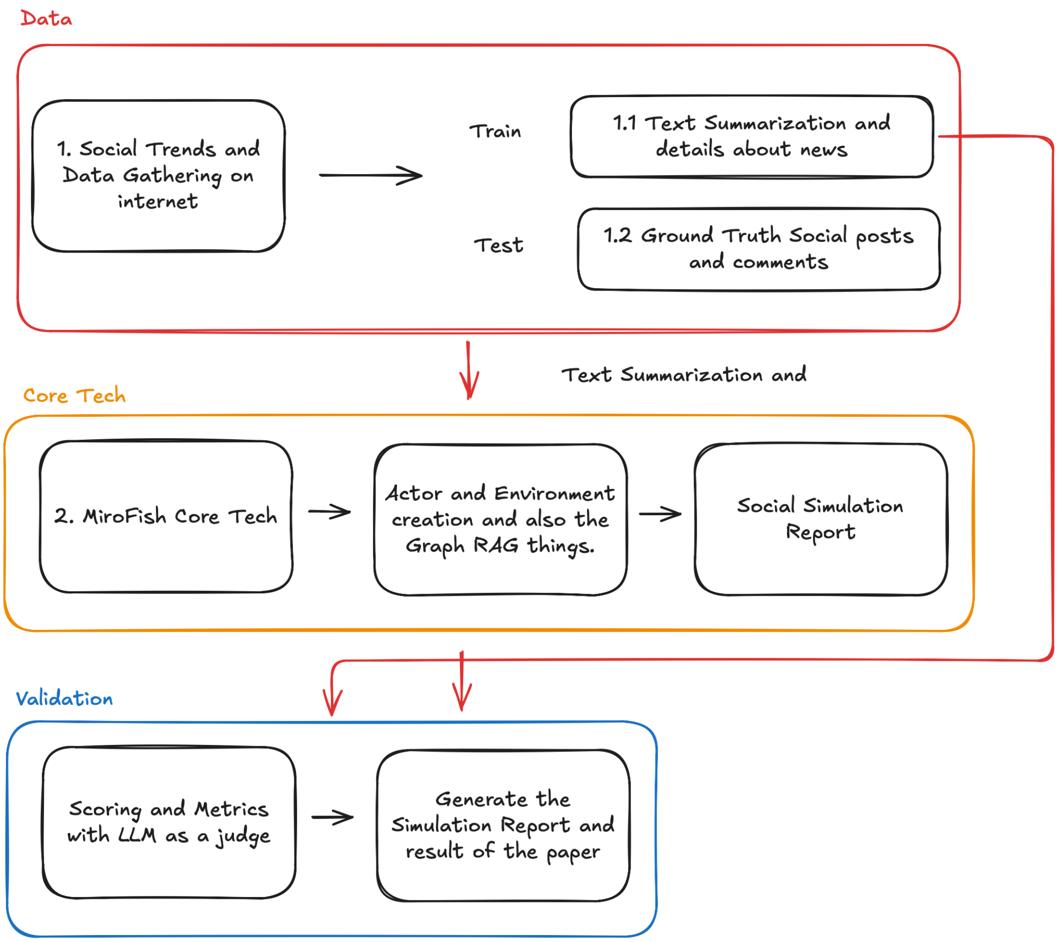

Fig. 1. DEEDY application architecture. Thai campaign context and social-media data are separated into a scenario-construction stream and a held-out validation stream. The MiroFish core builds a graph-grounded environment and actors, runs social simulation, and exposes outputs for multi-level evaluation.

version, complete run input, Apify run and dataset identifiers, start and completion times, requested limit (1,000), returned count, reply setting, error status, collection-code version, and a checksum of the frozen raw export. Raw UTF-8 JSONL is immutable after a successful run. Subsequent tables are derived from that snapshot and de-duplicated first by platform comment identifier and then by normalized-text similarity within a post. User names and platform identifiers are hashed or removed before research analysis.

This design improves repeatability but does not turn the result into a random sample. Search and post selection, platform ranking, deleted or moderated comments, public visibility, geography, login state, and Actor behavior can all affect what is returned. DEEDY therefore reports requested and observed counts for every post and treats cross-platform comparisons as conditional on these collection mechanisms.

4) Thai-Aware Data Views: The crawler preserves the Unicode comment text returned by each Actor in an immutable JSONL view. The implemented `convert_comments_to_csv.py` then flattens nested post records into a UTF-8-SIG table containing the campaign topic, comment, author, engagement, platform, post, comment and parent identifiers, and timestamp. This second view supports inspection and annotation without replacing the raw export.

DEEDY avoids aggressive text cleaning because emoji, Thai–English code-mixing, orthographic repetition, hashtags, particles, negation, and punctuation may carry campaign stance or affect. The analysis view standardizes only encoding, line breaks, and whitespace; any additional transformation is versioned and never overwrites the raw text. Thai tokenization is performed only when a metric requires tokens, while character- and embedding-level metrics are reported in parallel to reduce dependence on one segmentation system.

5) Evidence Split and Structured Weak Annotation: For each campaign e, the collected evidence is partitioned before any model summary is generated:

$$D_e = S_e \dot{\cup} G_e, \tag{2}$$

where S_e is the scenario-construction set and G_e is the held-out observed reaction set. S_e contains the campaign facts and context permitted at the chosen simulation start time. G_e contains comments and interactions that the simulator must reproduce but never receives as memory. A post is assigned wholly to one experimental role when post-level holdout is used; temporal experiments instead apply a pre-registered timestamp cut. Comment identifiers, quoted passages, and near duplicates cannot cross the boundary.

The frozen Apify records are passed to `llm_prelabel.py` in bounded batches. Three critique models independently assign a positive, neutral, or negative label to each sampled comment and summarize campaign response. A fourth model synthesizes their outputs into consensus sentiment labels, an overall sentiment-direction analysis, recommended campaign next steps, and possible business or operational changes. Exact sentiment counts are computed from the consolidated comment-level labels rather than estimated from the synthesis text. Additional evaluation labels for campaign stance, narrative, purchase intent, and sarcasm are collected under the human-annotation protocol rather than attributed to the current pre-labeling script.

These model outputs are *weak pre-annotations*, not ground truth. Only a summary derived from S_e may enter MiroFish; labels or summaries derived from G_e are restricted to scoring and error analysis. Before the main experiment, native Thai annotators review a frozen, source-stratified sample, adjudicate disagreements, report Krippendorff's α, and audit errors involving sarcasm, code-mixing, indirect criticism, and platform source. This separation prevents observed campaign reactions from leaking into agent memory.

C. MiroFish Core Engine

1) Reality Seed and GraphRAG: The scenario packet contains the campaign brief, creative message, dated factual claims, named entities, brands, products, creators, and source-level provenance from S_e. MiroFish extracts an ontology and entity relations, then injects individual and collective memories into its graph retrieval layer [11]. A retrieved memory attached to an agent action includes its supporting node or source identifier. Unsupported details generated during graph construction are marked as model assumptions and are not silently promoted to facts.

2) Actor and Environment Construction: Actors represent campaign-relevant audience roles justified by S_e, such as existing customers, prospective customers, creator followers, brand advocates, skeptics, journalists, community members, and bystanders. A profile separates evidence-backed attributes from sampled attributes. The Thai adaptation adds language register, Thai–English mixing rate, region or dialect only when supported, politeness strategy, typical message length, activity rate, campaign-interest prior, interests, and uncertainty. Demographic attributes are not inferred from a single message. Population weights and network links are sampled from documented priors and varied during sensitivity analysis rather than presented as known facts.

The environment specifies platforms, action space, time step, feed capacity, and recommendation mechanism. OASIS supplies posting, replying, reposting, following, and browsing behavior with dynamic social and content state [5]. DEEDY additionally records each exposure before an action. This makes it possible to distinguish "the agent never saw the message" from "the agent saw and rejected the message," a necessary distinction for interpreting diffusion failure.

3) Thai Context in Perception and Action: At each round, an actor retrieves relevant graph memory, observes a bounded feed, updates a private state, and chooses an action or abstention. Prompts require natural Thai consistent with the actor's register without demanding caricatured dialect. They preserve uncertainty, allow silence, and prohibit fabricated quotations or demographic claims. Conversation context is included for reply generation because Thai stance and sarcasm may depend on an earlier turn [17]. The language model is configurable: a Thai-optimized model such as Typhoon 2 and a strong multilingual model are compared under identical action rules rather than assuming either is superior.

4) Simulation Outputs and Application Layer: The core returns a machine-readable simulation log and a human-readable campaign report. The log contains agent state version, exposure set, retrieved evidence, selected action, generated content, parent action, timestamp, and token/model settings. The report summarizes dominant and minority narratives, sentiment and stance trajectories, influential nodes, propagation paths, disagreement, and unsupported claims. It also exposes uncertainty across runs.

The application allows a marketer or analyst to select a campaign, inspect the scenario packet and graph, configure actor count and feed policy, run a baseline or a counterfactual campaign message, compare it with held-out observations, and drill down from an aggregate chart to the simulated posts and evidence that produced it. Counterfactual reports compare conditions inside the model; they are not presented as causal effects in the real Thai consumer population.

IV. Experimental Design

A. Research Questions

The evaluation is organized around four questions:

- **RQ1 (Content):** Does DEEDY reproduce the sentiment, stance, narrative, and linguistic characteristics of held-out Thai reactions?
- **RQ2 (Population):** Does graph grounding and Thai adaptation improve aggregate correspondence without collapsing response diversity?
- **RQ3 (Dynamics):** Does the simulated interaction process match observed cascade size, shape, timing, and network structure?
- **RQ4 (Application):** Are campaign reports traceable, stable across repeated runs, and useful to Thai marketing analysts without overstating certainty?

B. Dataset and Splitting Protocol

The initial study will select campaigns with at least one usable factual scenario packet and multiple public campaign-related posts. Before simulation, the protocol freezes the required minimum returned-comment count, eligible platforms, post-source strata, and missing-data rules. Apify then receives a separate query for each post with a 1,000-comment limit as described in Section III-B2. Posts returning fewer records are retained when they satisfy the pre-registered minimum; the dataset never pads a short thread by duplicating comments or borrowing another post's quota.

The unit of generalization is a *campaign*, not an individual post or comment. Prompt and model selection use development campaigns, and final scores are reported on untouched test campaigns. Within a campaign, S_e and G_e are separated either by a pre-registered timestamp or by whole post so that a reaction thread cannot leak into both construction and evaluation. Actor run inputs, query dates, returned counts, source strata, and split assignments are frozen before weak annotation. This design supports source-balanced analysis, temporal-drift checks, and sensitivity to platform-specific public visibility.

Weak labels are independently reviewed by at least three native Thai annotators using a codebook for three-way sentiment, campaign-specific stance, sarcasm, and narrative membership. Annotators see necessary conversational context but not whether a post is real or simulated during quality comparison. Majority/adjudicated labels form the reference set, and Krippendorff's α is reported for each task. User identifiers are hashed and quoted examples are paraphrased unless publication is ethically justified.

C. Metrics

1) Content and Population Correspondence: Let P_G^y and P_S^y be observed and simulated distributions for label $y \in \{\text{sentiment}, \text{stance}, \text{narrative}\}$. Distributional correspondence is measured with Jensen–Shannon distance:

$$d_{\mathrm{JS}}(P_G^y, P_S^y) = \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P_G^y \| M) + \tfrac{1}{2} D_{\mathrm{KL}}(P_S^y \| M)}, \quad (3)$$

where $M = (P_G^y + P_S^y)/2$. Lower is better and the square-root form is bounded. Macro-F1 is also reported when simulated posts are matched to a conditioned stimulus or stance target.

Narrative correspondence is measured in two ways. First, Thai annotators map posts to a campaign-specific narrative codebook and we compare coverage and prevalence. Second, multilingual BERTScore measures semantic similarity between generated and observed response sets [19]. Embedding-based scores are treated as complementary because semantic proximity need not imply the same stance. Diversity is reported using Distinct-1/2, Self-BLEU, unique narrative count, lexical entropy, length distribution, and pairwise embedding distance. High correspondence accompanied by diversity collapse is considered a failure.

Polarization is measured from the stance distribution and interaction graph. We report normalized stance entropy, between-group versus within-group interaction, and modularity after fixing the community-detection procedure. Population statistics are computed per run and then aggregated across campaigns, so campaigns with many selected posts do not automatically dominate the result.

2) Temporal and Network Correspondence: For trace-enriched campaigns, cumulative activity curves are aligned to the campaign-post publication time. We report normalized root-mean-square error (NRMSE) and dynamic time warping distance, together with cascade scale, maximum depth, maximum breadth, structural virality, time to 50% of final activity, and peak-time error. OASIS uses scale, depth, breadth, and propagation-curve error in its own validation [5]; DEEDY adds exposure-conditioned action rates and Thai content labels.

Observed and simulated interaction graphs are compared by degree and cascade- size distributions using the two-sample Kolmogorov–Smirnov statistic, plus clustering coefficient, reciprocity, assortativity, modularity, and the share of cross-stance replies. These metrics are omitted, rather than imputed, when the source platform does not expose the required trace.

3) Thai Quality and Report Evaluation: Native Thai raters score blinded real and simulated posts on a five-point scale for naturalness, contextual coherence, stance consistency, pragmatic and cultural appropriateness, and persona consistency. Sarcasm precision, recall, and F1 are evaluated separately. Report claims are scored for evidence traceability: the proportion of factual claims linked to scenario evidence or simulation events, unsupported-claim rate, contradiction rate, and coverage of minority narratives.

An LLM judge may assist high-volume screening, but cannot be the sole quality measure. Pair order is swapped, output identity is hidden, the judge must cite the supplied evidence before scoring, and its agreement with Thai raters is reported. These controls address known position bias in LLM-as-a-judge evaluation [20].

D. Execution and Statistical Analysis

All stochastic conditions use ten seeds initially; a power analysis on pilot variance determines the final number. Model name and revision, prompts, temperature, top-p, agent count, network seed, recommendation parameters, round count, stopping rule, and token cost are frozen before the test set is opened. We report medians and 95% bootstrap confidence intervals across campaigns. Paired conditions use a campaign-level permutation test or Wilcoxon signed-rank test with Holm correction. Effect sizes accompany p-values.

The main comparison is the full DEEDY pipeline versus translated/unadapted MiroFish. Component ablations remove graph retrieval, Thai context features, thread context, or observed population weights one at a time. The deliberately leaky condition in Table I is never a valid system result; it is a diagnostic ceiling showing how much apparent performance can be gained by exposing held-out reactions.

E. Acceptance Criteria for the Application

Before results are observed, the study will set minimum criteria: (i) full pipeline completion and provenance for every scored run; (ii) statistically lower sentiment/stance Jensen–Shannon distance than the unadapted baseline; (iii) no significant diversity collapse relative to held-out posts; (iv) Thai quality above the midpoint with acceptable annotator agreement; (v) report unsupported-claim rate below a pre-registered threshold; and (vi), for trace-enriched campaigns, improvement on at least two cascade metrics without material

TABLE I

PLANNED EXPERIMENTS. EACH CONDITION IS RUN WITH THE SAME CAMPAIGN SPLIT, AGENT BUDGET, ROUND BUDGET, AND TEN RANDOM SEEDS UNLESS THAT FACTOR IS VARIED.

Experiment	Conditions	Primary purpose
Exp. 1: Grounding	DEEDY seed-only; ungrounded LLM agents; deliberately leaky all-source upper bound	Test whether held-out correspondence comes from the scenario pipeline rather than parametric knowledge or target leakage.
Exp. 2: Thai adaptation	Full Thai adaptation; translated/unadapted MiroFish; normalization without context features	Measure sentiment/stance distance, narrative coverage, Thai quality, sarcasm, code-mixing, and diversity.
Exp. 3: Core model	Thai-optimized LLM; matched multilingual LLM; simple non-generative ABM baseline	Separate the contribution of the language model from the graph, population, and platform mechanisms.
Exp. 4: Social mechanism	Chronological, popularity, and interest feeds; observed versus random network; multiple population sizes	Test cascade, polarization, herding, and sensitivity to platform assumptions.
Exp. 5: Application	Baseline campaign versus one pre-registered message-framing counterfactual; report with versus without uncertainty/evidence links	Assess traceability, analyst usefulness, and whether presentation changes trust.

degradation on the others. Failure remains a reportable result and defines the application's operating boundary.

V. LIMITATIONS AND ETHICAL CONSIDERATIONS

The campaign corpus is a convenience sample of manually selected public posts from Facebook, Instagram, and TikTok and cannot represent the Thai population or each platform's users. Apify Actors can retrieve only comments exposed by their collection mechanism; the 1,000-comment request may therefore return fewer records and may reflect ranking, geography, login state, moderation, deletion, platform change, or Actor failure. Coordinated activity and bots may also bias observed response. Requested and returned counts, Actor versions, and failures must consequently accompany every result. Even lightweight normalization can alter pragmatic cues, so linguistic claims must be checked against the paired raw view.

Personas can reproduce stereotypes even when generated from real traces. DEEDY therefore avoids inferring sensitive attributes from isolated posts, reports subgroup results only where data and ethics permit, and presents regional or dialect attributes as uncertain rather than essential. Public availability does not remove privacy risk: identifiers must be hashed, raw text access restricted, quotations minimized, and collection reviewed against platform terms and institutional research-ethics requirements.

Finally, LLM agents share training data, architecture, and prompt conventions; agreement between them may reflect a common model prior rather than social consensus. Counterfactual campaign findings are conditional on the scenario, actor sampling, network, recommender, and model version. DEEDY is therefore an instrument for exploratory comparison and hypothesis generation, not a replacement for campaign experiments, customer research, surveys, or participatory research with Thai communities.

VI. CONCLUSION AND FUTURE WORK

Placeholder—to be completed after the data pipeline, MiroFish core, and experiments are finalized. The final conclusion will summarize measured findings, supported application claims, failure cases, and next steps without introducing results that are not reported in the paper.

REFERENCES

[1] C. M. Macal and M. J. North, "Tutorial on agent-based modelling and simulation," in *Proceedings of the 2010 Winter Simulation Conference*, 2010, pp. 151–162.

[2] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–22.

[3] J. S. Park, L. Popowski, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Social simulacra: Creating populated prototypes for social computing systems," in *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 2022, pp. 1–18.

[4] C. Gao, X. Lan, Z. Lu, J. Mao, J. Piao, H. Wang, D. Jin, and Y. Li, "S^3: Social-Network Simulation System with Large Language Model-Empowered Agents," 2023.

[5] Z. Yang, Z. Zhang, Z. Zheng, Y. Jiang, Z. Gan, Z. Wang, Z. Ling, J. Chen, M. Ma, B. Dong, P. Gupta, S. Hu, Z. Yin, G. Li, X. Jia, L. Wang, B. Ghanem, H. Lu, C. Lu, W. Ouyang, Y. Qiao, P. Torr, and J. Shao, "OASIS: Open agent social interaction simulations with one million agents," in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: https://arxiv.org/abs/2411.11581

[6] J. Tang, H. Gao, X. Pan, L. Wang, H. Tan, D. Gao, Y. Chen, X. Chen, Y. Lin, Y. Li, B. Ding, J. Zhou, J. Wang, and J.-R. Wen, "GenSim: A general social simulation platform with large language model based agents," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, 2025, pp. 143–150. [Online]. Available: https://aclanthology.org/2025.naacl-demo.15/

[7] R. Arreerard, S. Mander, and S. Piao, "Survey on Thai NLP language resources and tools," in *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. European Language Resources Association, 2022, pp. 6495–6505. [Online]. Available: https://aclanthology.org/2022.lrec-1.697/

[8] D. Kim, S. Lee, Y. Kim, A. Rutherford, and C. Park, "Representing the under-represented: Cultural and core capability benchmarks for developing thai large language models," in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 4114–4129. [Online]. Available: https://aclanthology.org/2025.coling-main.278/

[9] J. Mooney, J. Woldense, Z. R. Jia, S. A. Hayati, M. H. Nguyen, V. Raheja, and D. Kang, "Are LLM agents behaviorally coherent? latent profiles for social simulation," 2025.

[10] N. Schwager, S. Münker, A. Plum, and A. Rettinger, "Towards simulating social media users with LLMs: Evaluating the operational validity of conditioned comment prediction," in *Proceedings of the 15th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 2026, pp. 208–221. [Online]. Available: https://aclanthology.org/2026.wassa-1.16/

[11] MiroFish Contributors, "MiroFish: A simple and universal swarm intelligence engine," Open-source software repository, 2026, accessed 12 August 2026. [Online]. Available: https://github.com/666ghj/MiroFish

[12] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate, "Out of one, many: Using language models to simulate human samples," *Political Analysis*, vol. 31, no. 3, pp. 337–351, 2023.

[13] L. Cross, N. Haber, and D. L. K. Yamins, "Validating generative agent-based models of social norm enforcement: From replication to novel predictions," 2025.

[14] W. Phatthiyaphaibun, K. Chaovavanich, C. Polpanumas, A. Suriyawongkul, L. Lowphansirikul, P. Chormai, P. Limkonchotiwat, T. Suntorntip, and C. Udomcharoenchaikit, "PyThaiNLP: Thai natural language processing in python," in *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software*, 2023, pp. 25–36. [Online]. Available: https://aclanthology.org/2023.nlposs-1.4/

[15] L. Lowphansirikul, C. Polpanumas, A. Rutherford, and S. Nutanong, "WangchanBERTa: Pretraining transformer-based thai language models," 2021.

[16] K. Pipatanakul, P. Manakul, N. Nitarach, W. Sirichotedumrong, S. Nonesung, T. Jaknamon, P. Pengpun, P. Taveekitworachai, A. Na-Thalang, S. Sripaisarnmongkol, K. Jirayoot, and K. Tharnpipitchai, "Typhoon 2: A family of open text and multimodal thai large language models," 2024.

[17] D. Hazarika, S. Poria, S. Gorantla, E. Cambria, R. Zimmermann, and R. Mihalcea, "CASCADE: Contextual sarcasm detection in online discussion forums," in *Proceedings of the 27th International Conference on Computational Linguistics*, 2018, pp. 1837–1848. [Online]. Available: https://aclanthology.org/C18-1156/

[18] Apify, "Actors documentation: Runs and builds," Online documentation, 2026, accessed 12 August 2026. [Online]. Available: https://docs.apify.com/actors/running/runs-and-builds

[19] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=SkeHuCVFDr

[20] L. Shi, C. Ma, W. Liang, W. Ma, and S. Vosoughi, "Judging the judges: A systematic study of position bias in LLM-as-a-judge," 2024.